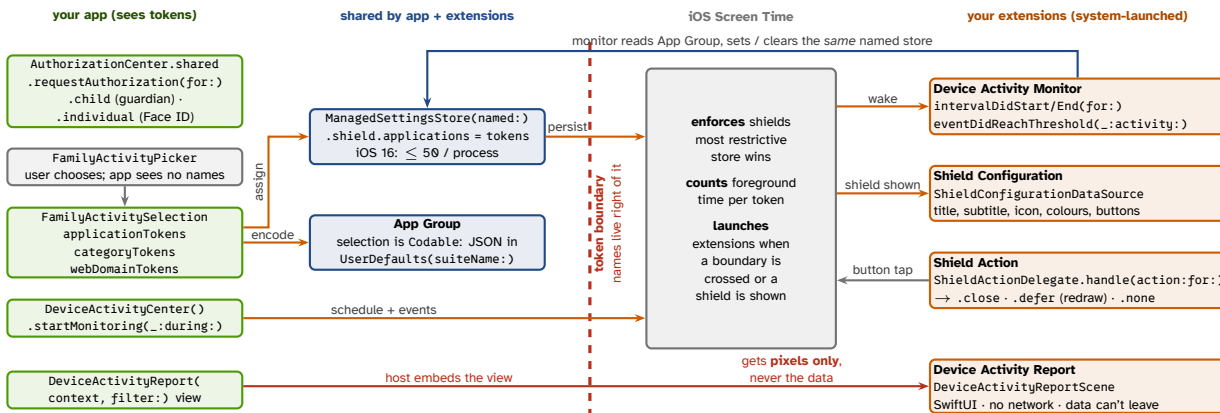


**In one line:** Three frameworks, one privacy trick. **FamilyControls** authorizes the app and lets the user pick apps/sites in a system picker that hands back **opaque tokens** — the app **never learns which apps**. **ManagedSettings** puts tokens into a **store** the system enforces (shields, restrictions). **DeviceActivity** wakes a **monitor extension** on schedules and usage thresholds. Usage data is shown only by a **sandboxed report extension**.



## How it works

- Authorize** (iOS 16+, async throws). `.child: iCloud Family Sharing child`; a guardian approves on the child's device (Apple ID + password) → the app **can't be deleted** without them, iCloud sign-out blocked. `.individual: Face ID / Touch ID / passcode`, **no lock**, many apps per device. Status: `.notDetermined` / `.denied` / `.approved`.
- Tokens:** `Token<T> = ApplicationToken, ActivityCategoryToken, WebDomainToken` — Codable, Hashable, no name inside. `SwiftUI Label(token):` the system draws icon + name, your code can't read it. **Revoking voids them all.**
- Store groups:** shield, application (denyAppRemoval), appStore, account, media, webContent, passcode, cellular... nil removes your value. Settings **persist without your process.**
- Shield:** applications, webDomains (token sets); `applicationCategories, webDomainCategories: .specific(_except:) · .all(except:) · .none`.
- Schedule:** `DeviceActivitySchedule` (start/end `DateComponents`, repeats, `warningTime`). **Event:** token sets + threshold of cumulative **foreground** time (empty sets = all); includes `PastActivity` iOS 17.4. `warningTime` adds `intervalWillStart/EndWarning`, `eventWillReachThresholdWarning`. Throws `MonitoringError(.excessiveActivities, .intervalTooShort...)`.

## Example — allow social apps only 17:00–20:00

```
// App: after .familyActivityPicker(isPresented:selection:)
let social = ManagedSettingsStore(named: .init("social"))
social.shield.applications = sel.applicationTokens
social.shield.applicationCategories = .specific(sel.categoryTokens)
group.set(try JSONEncoder().encode(sel), forKey: "social") // App Group
try DeviceActivityCenter().startMonitoring(.init("evening"),
    during: DeviceActivitySchedule(intervalStart: DateComponents(hour: 17),
    intervalEnd: DateComponents(hour: 20), repeats: true))
// Device Activity Monitor extension (principal class)
class Monitor: DeviceActivityMonitor {
    override func intervalDidStart(for a: DeviceActivityName) {
        ManagedSettingsStore(named: .init("social")).clearAllSettings()
    }
    override func intervalDidEnd(for a: DeviceActivityName) {
        // decode the selection from the App Group, re-assign the shield
    }
}
```

## Entitlement, setup

`com.apple.developer.family-controls` on the app and each extension. Adding the capability covers **development**; for **distribution** the Account Holder requests it (request form or the *Capability Requests* tab) — once per extension type too: Monitor, Report, Shield Action, Shield Configuration. Apple grants it as a **managed capability**. One App Group on every target.

## New in iOS 26.x (docs)

**26.4:** `shield.secondaryButtonSubmenuItems` → `.firstSecondarySubmenuItemPressed ... third...` **26.4, EU only:** `.approvedWithDataAccess + FamilyActivityData` (real bundle ids, visited domains) with `...family-controls.app-and-website-usage`; **one app per device**; elsewhere `FamilyControlsError.unavailable`. **26.5:** `ShieldActionResponse.openParentalControlsApp`.

## The four extensions

|                       |                                                                                                                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Monitor</b>        | subclass <code>DeviceActivityMonitor</code> . Runs when <b>no app does</b> (the child never opens yours). Short-lived, tight memory: read App Group, write stores, return.                                           |
| <b>Shield Config.</b> | <code>ShieldConfigurationDataSource.configuration(shielding:)</code> per Application / WebDomain (+ in: category) → <code>ShieldConfiguration</code> . Too slow → default look.                                      |
| <b>Shield Action</b>  | <code>ShieldActionDelegate: .primaryButtonPressed / .secondaryButtonPressed + a token</code> → <code>completion(ShieldActionResponse)</code> .                                                                       |
| <b>Report</b>         | <code>DeviceActivityReportExtension</code> + scenes; <code>makeConfiguration(representing:)</code> async gets <code>DeviceActivityResult</code> s; host picks scene by Context + <code>DeviceActivityFilter</code> . |

## Traps

- ! “Get the list of apps the user picked” — you can't (outside 26.4 EU data access): tokens only. Persist the **selection**, not names.
- ! `intervalDidStart` is not an alarm: it fires at the **first device use** after the boundary. No use, no callback.
- ! Set in the app, cleared in the extension? The **same named store** — shared by app + extensions (iOS 16).
- ! Clearing one store doesn't unblock what another store shields: **most restrictive wins**.
- ! Development builds work without Apple's approval; distribution needs it for the app and each extension type — request it early.

## Remember

**“Pick → token → store → system enforces; extensions react; the report paints, the app never reads.”** FamilyControls = *who may*, ManagedSettings = *what is blocked*, DeviceActivity = *when*.